

Problem Validation



Engineering Design Project

University Of Moratuwa

Automatic Streetlight Control and Fault Detection System

This questionnaire is intended to understand the current streetlight operation, existing problems, and the necessity of an automated fault detection and control system within the municipal council area.

1. What type of streetlights are mainly used in your area?

- a. LED
- b. Sodium Vapor
- c. CFL
- d. Mixed

They are using CFL at the moment in most of the inside streets but in the main streets they are using Sodium vapor recently the municipal council is changing CFL to LEDs

2. How streetlights are currently switched ON and OFF?

- a. Manuel Switching
- b. Time-based system
- c. Light sensor (LDR)
- d. Centralized Control

99 percent if the lights are manuel but they tested timebased systems the test is on going

3. How often do you observe streetlight remaining ON during daytime?

- a. Frequently
- b. Occasionally
- c. Rarely
- d. Never

The manuel switching means the switch sometimes connected to the council or it will be in the end of the street the locals have to on and off they see occasionally the street light remaining on in the day time

4. How often do streetlight fault occur in your area?

- a. Very often
- b. Often
- c. Rarely
- d. Very Rarely

Often happen with the CFL and bat problems

5. What are the common types of faults experienced?

- a. Bulb failure
- b. Power supply failure
- c. Wiring damage
- d. Control unit failure
- e. Other:-

Bulb failure and wiring damage

6. How streetlight faults are usually detected?

- a. Public complaints
- b. Routine inspection
- c. Maintenance staff observation
- d. No fixed method

Mostly Public complaints but sometimes we do routine checkups

7. On average, how long does it take to identify a faulty streetlight?

- a. Within 24 hours 1 to 3 days
- b. 1–3 days
- c. More than 3 days

8. What problems arise due to delayed fault detection?

- a. Increased energy wastage Public safety concerns
- b. Public safety concerns
- c. Increased maintenance cost
- d. Public complaints

9. Is energy wastage due to streetlights a concern for the council?

- a. Yes yes
- b. No
- c. Not Sure

10. Would an automatic ON/OFF control system be beneficial?

- a. Yes yes
- b. No they assigned a particular man to on and off the main street lights and they are paying them if this system comes it will reduce their expenditure
- c. Maybe

11. Would a system that automatically detects and reports streetlight faults be useful?

- a. Very Useful Yes
- b. Useful
- c. Not Useful

12. What are the main concerns regarding such a system?

- a. Installation cost
- b. Maintenance complexity The system has to be easily understandable to our staffs and the installation cost has to be low
- c. Reliability
- d. Safety
- e. Other:-

13. Do you have any suggestions or comments regarding improving streetlight management?

